

AUTOMATION OF MEASUREMENTS OF SIGNALS OF AIRCRAFT TRANSPONDERS IN INFORMATION MEASURING STANDS ON THE BASIS OF STANDARD VP AND SDR TECHNOLOGIES

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Currently, to solve measurement problems, measurement automation, there is an established set of measurement standards and platforms, as well as a set of software development tools. At the same time, it should be noted that the use of the open measuring standard PXI in conjunction with the graphical programming environment LabVIEW is currently not uncommon and is increasingly used in various fields [1, 2], including aircraft instrumentation [3]. At the same time, one of the most innovative approaches to building automated PXI/PXIe test benches is to use measurement modules based on programmable radio technology (software-defined radio, SDR) [4, 5], which greatly simplifies the programming and debugging process by using LabVIEW unified software development environment.

The basic rules for transmitting request signals and receiving aircraft response signals in a secondary radar system are considered. The main components of automated test benches for checking aircraft transponders are considered. The main methods for measuring the energy and time characteristics of response signals are considered. The main functions of the measuring stand, built on the basis of the measuring standard PXI and SDR, are shown, its interaction with the aircraft transponder by means of protocols based on ARINC 664 Part 7 is shown. The main categories of checks of aircraft transponders are considered. It is noted that the use of traditional test equipment is undesirable due to the likely operator overload. It is shown that currently the AFDX networks are widely used in the design of onboard surveillance systems. It is noted that their inclusion in the contour of the automated measuring stand allows to increase the automation of checks.

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Introduction

Currently, LabVIEW is not only an effective software development tool for measuring systems based on traditional PXI measurement modules, but also for PXI modules that use built-in high-speed FPGAs to solve measurement problems, which are the basis of SDR. Thus, the basic parameters of the hardware components of the automated measuring stands will be reconfigurable (that is, programmable), and not hardware-dependent, as in traditional analogue transceiver systems.

This approach provides several advantages:

- the possibility of reconfiguration of the FPGA in the automated measuring stand, depending on the type and purpose of the on-board unit being checked;
- the possibility of synthesizing almost any measuring instrument developed for specific narrowly focused measuring tasks;
- the possibility of analyzing and researching rapid processes associated with the temporal characteristics of radio transceiver cycles;
- relative simplicity of integration into already existing measuring systems and complexes.

In particular, this approach is used in the construction of information measuring stands for testing surveillance systems [6].

Secondary location tasks

We give a brief description of the surveillance system. The basis of the observation system is a secondary location system whose operation boils down to the following: a beam of a narrowly directed antenna rotating 360° through pulses of a certain shape and sequence generates so-called. request [7], which is transmitted from the terrestrial interrogator to the aircraft's transponder at a frequency of 1030 MHz. Aircraft transponders receiving this request process it and transmit the response information to the ground station at a frequency of 1090 MHz, also in the form of a specific form and sequence of pulses (fig. 1).

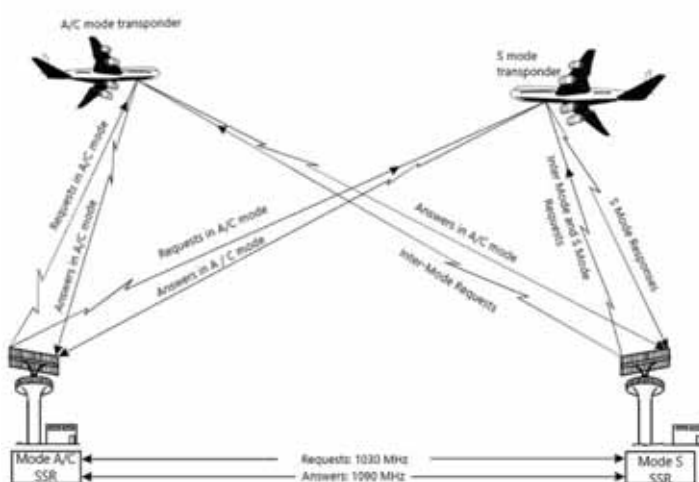


Fig. 1. General scheme of functioning of SSR

In civil aviation, airborne transponders operate in the following modes: A/C modes ATCRBS (Air Traffic Control Radar Beacon System) and Mode Select (Mode S), which is currently the main.

The basis of the request signal of mode S is a pulse-modulated signal with internal phase modulation. Consider it in detail. Mode S request consists of three main pulses: P1, P2 and P6. Before pulse P6, a pair of pulses P1 – P2 is transmitted, which suppresses responses from transponders with A/C mode in order to eliminate synchronous distortions due to accidental operation from requests in mode S. Pulse P6 is phase-modulated. The very first phase change at 180° is called synchronous phase reversal. Synchronous phase reversal in pulse P6 indicates the point in time from which the demodulation of time intervals (so-called data chips) with a duration of $0.25 \mu\text{s}$ begins. This series of data chips begins $0.5 \mu\text{s}$ after synchronous phase reversal and ends $0.5 \mu\text{s}$ before the falling edge of pulse P6 (fig. 2). The phase reversal may or may not take place in front of each data chip for encoding, the value of binary information reflected by it. Phase reversal during data transmission takes place only at a time equal to $N \times 0.25 \mu\text{s} + 0.02 \mu\text{s}$ (N is equal to or greater than 2) after synchronous phase reversing. A pulse of P6 with a duration of $16.25 \mu\text{s}$ contains no more than 56 rollovers of the data phase. Pulse P6 with a duration of $30.25 \mu\text{s}$ contains no more than 112 phase overturns during data transmission.

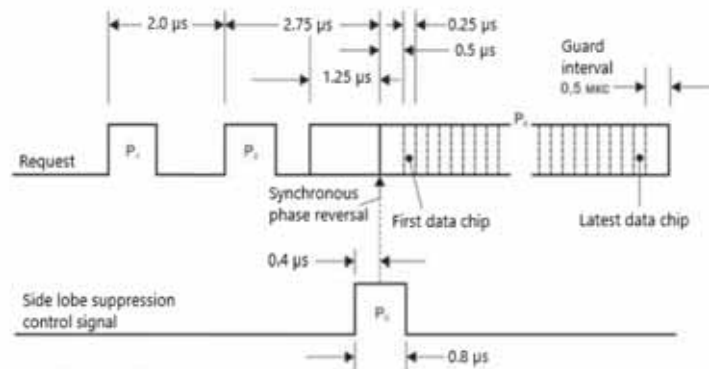


Fig. 2. Structure of the request signal mode S

The mode S response signal is based on a PPM modulated signal, the information in which is encoded by means of the position of the pulses.

In this case, it is a signal consisting of four preamble pulses (representing the entry into synchronism) and 56 or 112 one microsecond segments, where a pulse of $0.5 \mu\text{s}$ is present either in its first or in the second half one microsecond interval [7]. The pulse contained in the first part of one microsecond segment is a binary ONE, and if the pulse is contained in the second part one microsecond segment, then this is a binary ZERO. The numbering of bits occurs in the order of their transmission, that is, it starts from the first bit, where the first bit transmitted in the message is the most significant bit (MSB) [7]. Information is encoded in fields, each of which consists of at least one bit. It should be noted that this mode is also used by other systems working in conjunction with the bota transponder, for example, the AZN-B system (Automatic Dependent Surveillance System in the Broadcast Mode). The structure of S mode responses is shown in figure 2.

As it is easy to assume from fig. 2 and 3, requests and responses of mode S for the purpose of unambiguous decoding is a deterministic set of bit fields containing a certain type of request and response in the form of a certain type of information

about the aircraft. We list the main request formats (UF) and responses (DF) used by SSR and give them in table 1.

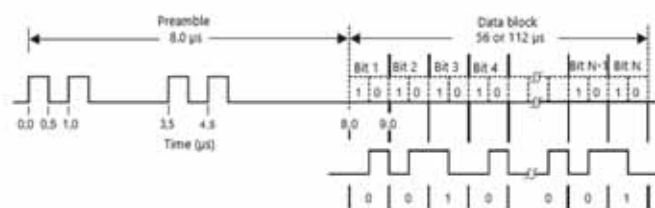


Fig. 3. Structure of the response signal mode S

Table 1

The main formats of requests and responses used by SSR

Request format	Response format
UF = 4 – observation, request height data	DF = 4 – observation, response with height data
UF = 5 – observation, identification request	DF = 5 – observation, identification response
UF = 11 – general call in S mode only	DF = 11 – general call request
UF = 20 – height data request, Comm-A	DF = 20 – height response, Comm-B
UF = 21 – identification request, Comm-A	DF = 21 – identification response, Comm-B

The task of checking aircraft transponders

The specificity of checking aircraft transponders is that in the process of checking them, a sufficiently large number of checked parameters are used: starting from the time parameters of the signals and ending with the time of issuing the response to the request. In addition, one of the main types of checks are the physical parameters of the signals, both received by the transponder and transmitted by it.

A complete list of parameters to be verified is set out in DO-181E "Minimum operational air traffic control radar beacon system/mode select (ATCRBS/mode S) airborne equipment". Highlight the main categories of checks:

A. Configuring requests

- configuration of physical parameters of requests;
- configuration of temporary parameters of requests;
- configuring information characteristics of requests;
- configuration of quantitative parameters of requests;

B. Control of accepted responses

- control of the physical characteristics of response signals;
- control of time characteristics of received responses;
- control of the quantitative characteristics of the received responses;
- control of information characteristics of the received answers.

An analysis of the types of inspections of aircraft transponders set out in DO 181E [8] showed that:

a) DO-181E testing procedures involve the use of various traditional measuring instruments or test equipment (CPS), such as TIC-48 or their equivalent, signal analyzers, etc.;

b) To implement the functionality of checks in automated test benches, it is advisable to use reconfigurable transceivers based on FPGA (Field Programmable Gate Array). In practice, one of these options is the use of programmable FPGA arrays as part of the FlexRIO 7966R digital modulator/demodulator and a transceiver module, such as the NI 5791R.

A typical scheme for checking aircraft transponders using traditional KPA is shown in fig. 4.

As can be seen from fig. 4, the operator is entrusted with many functions associated with performing an inspection and requiring him to increase concentration and attention. Given the complexity and diversity of the checks performed, their scenarios, and, moreover, the variety of auxiliary work performed by the operator, for example, recording measurement results in the protocol, performing various mechanical routine work, we should talk about its overload, which objectively reduces the speed of testing, repeatability measurements, and also increases the likelihood of erroneous actions due to his fatigue.

Increased automation of verification through the use of AFDX

It is worth noting that there is already a trend towards the transition from on-board exchange networks, built on the basis of point-to-point channels and multiple access channels with centralized control, to switched networks based on the ARINC 664 standard «Aircraft Data Network. Part 7. AVIONICS FULL-DUPLEX SWITCHED ETHERNET NETWORK». ARINC 664 is a standard that defines the protocol and electrical connection specifications for information networks used in avionics.

The main purpose of the ARINC 664 standard is to create a deterministic data transmission network that can be used for the systems necessary for flight control. This goal is achieved by providing dedicated traffic bandwidths for each information route in the network and ensuring the availability of the Quality of Service (QoS) specification at each system node [9, 10].

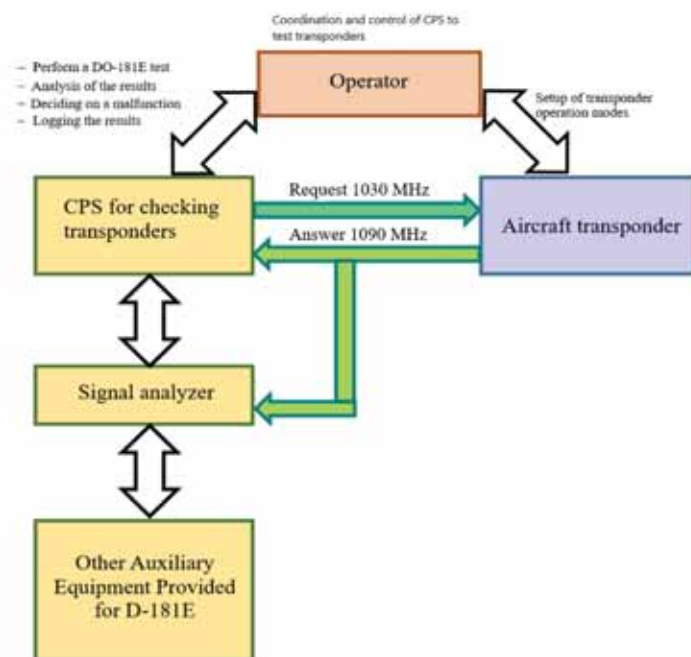


Fig. 4. Typical test scheme for aircraft transponders

Thus, in automated test benches, it is possible to extend the functionality and automate the verification by using AFDX to control the operating modes of the transponder, as well as to simulate the input data necessary for their operation.

The inclusion of AFDX in the circuit of the prototype of the stand allows test procedures running on the automated workplace of the operator (AWS) to control the verification process of the transponder, which is the object of control (OK) (fig. 5).

Taking into account the scheme shown in fig. 4, the operator selects one or several OK tests through the special software (OSS), after which the workstation through the TCP/IP protocol stack configures the stand for the selected test type, and also configures the OK via AFDX protocol: sets the required operation mode and provides other necessary information provided by the test program. The bench sends a test signal, consisting generally of pulses that form the required request mode, receives a response signal, analyzes the required parameters provided by the test program and transmits the results to the AWS via TCP/IP, and itself goes into standby mode until the beginning of the next test. The results of the analysis obtained by ARM are compared with the required standard, after which the next test is performed. Upon completion of all the tests selected by the operator, a protocol with test results is generated, stored and displayed on the display facilities.

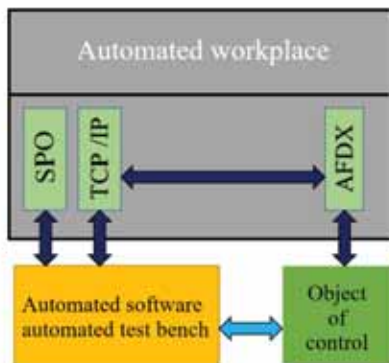


Fig. 5. The general scheme of the automated measuring stand using AFDX

Based on the scheme shown in fig. 5, as well as the description of the verification process, the block diagram of the automated measuring stand is shown in fig. 6.

Given the above scheme, the verification process OK is as follows. Using RF cables, the stand is connected to the OK, after which the operator launches a top-level program on the HOST PC, the so-called. test manager, and selects one or more test programs.

Before the start of each test, the test program, if necessary, configures the OK via AFDX, selects the required mode of its operation, and then generates the type of testing signal and the required number of transceiver cycles, and then sends the given signal to OK. An FPGA-based SDR receiver consisting of a booth receives and demodulates signals received from the OC, measures the parameters provided by the test and sends them (if necessary) to the real-time system for decoding. The results of the measured parameters are returned to the test manager, where the test results from the measurement protocol.

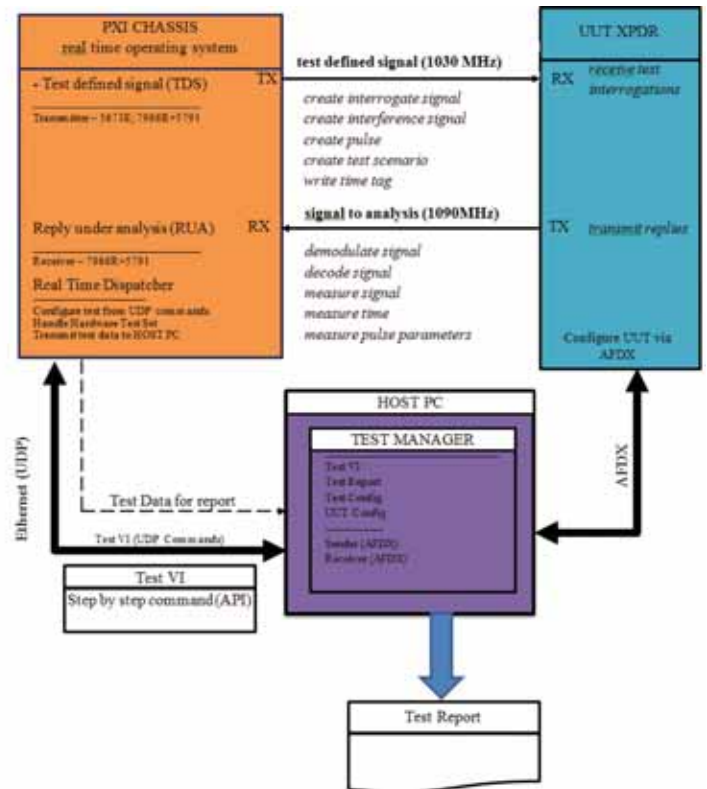


Fig. 6. Structural automated measuring bench

A test program is a file containing a set of commands that form the verification algorithm. A test program is an API function that generates and transmits control commands via a TCP/IP protocol stack to a network packet manager in a real-time system. The network packet manager in the real-time system receives incoming packets, filters and distributes them to the appropriate subroutines responsible for the operation of the corresponding hardware. Each subroutine accepts commands that are intended only for it and translates them into low-level hardware control commands, receives data from the hardware, translates them into network packets, and sends them back to the packet manager, which then transfers them to the test program.

The architecture of open source software is shown in fig. 7.

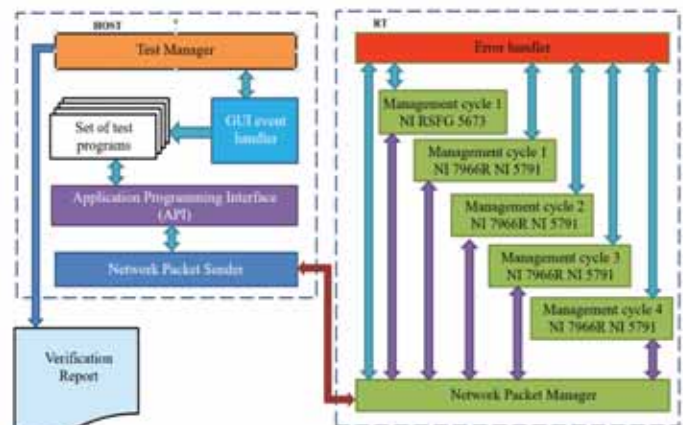


Fig. 7. Special software architecture

Conclusion

Thus, the inclusion of AFDX networks in the circuit of an automated measuring bench allowed us to automate the following processes related to the verification of aircraft transponders:

- execution of test procedures associated with the change in the physical parameters of signals in time;
- execution of test procedures associated with the change in the quantitative parameters of signals in time;
- counting the received answers and decoding them;
- setting up the modes of operation of aircraft transponders;
- recording of measurement results in the protocol and comparing them with the norm;
- taking into account the accumulated statistics for each verified transponder, it is possible to predict their technical condition.

The analysis of DO-181E, as well as the stages of work on the verification of aircraft transponders, allows us to conclude that:

- the combined use of virtual instrument technology and SDR reduces the development time of new and adaptation of existing automated measurement stands and systems through a single development environment;
- to increase the functionality of the automated measuring stand through the use of reconfigurable measuring modules based on PXI/PXIe.

Using the capabilities of AFDX in the design of automated booths provides a number of the following advantages:

- increase the rate of automation of verification procedures by using the AFDX protocol to control the OK;
- the ability to analyze and explore the rapid processes associated with the transmission and reception of aviation signals, through the use of modular high-speed vector transceivers based on FPGA.

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АВТОМАТИЗАЦИЯ ИЗМЕРЕНИЙ СИГНАЛОВ САМОЛЁТНЫХ ПРИЕМООТВЕТЧИКОВ В ИНФОРМАЦИОННЫХ ИЗМЕРИТЕЛЬНЫХ СТЕНДАХ НА ОСНОВЕ СТАНДАРТНЫХ ТЕХНОЛОГИЙ ВП И SDR

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Аннотация

Рассмотрены основные правила передачи сигналов запроса и приема сигналов ответа воздушных судов в системе вторичной радиолокации. Описаны компоненты автоматизированных измерительных стендов для проверки приемоответчиков воздушных судов. Рассмотрены основные способы измерения энергетических и временных характеристик ответных сигналов. Приведены функции измерительного стенда, построенного на основе измерительного стандарта PXI и SDR, показано его взаимодействие с самолетным приемоответчиком посредством протоколов на основе спецификации ARINC 664 часть 7. Рассмотрены основные категории проверок самолетных приемоответчиков. Отмечено, что использование традиционной контрольно-проверочной аппаратуры нежелательно, ввиду вероятной перегрузки оператора. Показано, что в настоящее время получило широкое применение сетей AFDX при проектировании бортовых систем наблюдения. Сформулирован вывод, что включение этих сетей в контур автоматизированного измерительного стенда позволяет повысить автоматизацию проверок.

Ключевые слова: виртуальный прибор, виртуальная измерительная система, информационно-измерительная система, AFDX, FPGA, SDR, LabVIEW, стандарт PXI.

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